de Hevesy and Chiewitz in 1935 synthesized ^{32}P (β^- $t_{½}$ 14.3 d) and used this tracer in biological studies. In the same year de Hevesy and co-workers also carried out activation analyses on rare earths. Despite the demonstration of the value of the tracer technique by these early studies the technique did not come into common use until after World War II when relatively large amounts of cheap radionuclides became available through the use of nuclear reactors.

While it is not necessary to use radioactive isotopes for tracer studies, in general, the use of radioactivity is simpler and less expensive than the use of stable isotopes. Research with the latter requires rather sophisticated and expensive measuring devices such as mass spectrometers, cf. §3.3.2. We restrict our discussion to the use of radioactive tracers.

Among the advantages of using radiotracers we can list the following: (a) radiotracers are easy to detect and measure with high precision to sensitivities of 10^{-16} to 10^{-6} g; (b) the radioactivity is independent of pressure, temperature, chemical and physical state; (c) radiotracers do not affect the system and can be used in nondestructive techniques; (d) if the tracer is radiochemically pure, interference from other elements is of no concern (common in ordinary chemical analyses); (e) for most radioisotopes the radiation can be measured independently of the matrix, eliminating the need for calibration curves.

18.1. BASIC ASSUMPTIONS FOR TRACER USE

In some experiments answers to scientific questions which require knowledge of the presence and concentration of a specific element or compound at a certain place and at a certain time can be obtained only through the use of a radioactive tracer. For example, self diffusion of metal ions in solutions of their salts cannot easily be studied by any other technique. However, in other cases the use of radioactive tracers is not necessary in principle but is justified by the greater convenience. In either type of investigation there are two assumptions implicit in such uses.

The primary assumption is that radioactive isotopes are chemically identical with stable isotopes of the same element, i.e. the substitution of ^{14}C for ^{12}C in a compound of carbon does not change the type or strength of the chemical bonds nor does it affect the physical properties of the compound. The validity of this assumption depends on the precision of measurement of the chemical and physical properties. The difference in mass between the various isotopes does cause some change in these properties (§3.5) but even in the case of ^{14}C and ^{12}C, with a mass difference of approximately 15%, the isotope effect is rather small and difficult to detect. Normally only for systems involving hydrogen-deuterium-tritium substitution must isotope effects be considered. For heavier elements it can be neglected in almost every situation.

The second assumption in the use of tracer techniques is that the radioactive nature of the isotope does not change the chemical and physical properties. Until the moment of

its disintegration the radioactive atom is indistinguishable from its stable isotope except for the isotopic mass difference. When the radioactive disintegration of the atom has been observed ("counted"), the decay product is normally a different element and its subsequent chemical behavior is usually of no interest. If the disintegration rate is very high, the energy released by the radioactive decay can cause observable secondary radiolytic effects (Ch. 8). However, in well-designed tracer experiments the level of radioactivity is high enough to provide accurate data but normally small enough not to produce noticeable chemical effects.

While the radioactivity of the tracers is assumed not to affect the chemical systems, the parent-daughter relationship of radioactive nuclides needs special consideration. For example, since strontium and yttrium are not chemically identical, a gross β-count of strontium samples including ^{90}Sr may include an unknown fraction of ^{90}Y activity present from ^{90}Sr decay because of the relationship

$$^{90}\mathrm{Sr}(\beta^-, t_{½}\ 28.5\ \mathrm{y})\ ^{90}\mathrm{Y}\ (\beta^-\gamma, t_{½}\ 2.671\ \mathrm{d})\ ^{90}\mathrm{Zr}(\mathrm{stable}) \qquad (18.1)$$

Beta-absorption, and β- or γ-scintillation techniques which use energy discrimination, are frequently useful in such parent-daughter cases. If equilibrium is rapidly established between the parent and daughter activities it is usually simpler to count the samples after sufficient time for this to occur, the contribution to the observed count rate by the daughter is then proportional to the amount of mother in the sample. In the case of ^{90}Sr—^{90}Y, radioactive equilibrium is established in about 25 d. If ^{137}Cs ($\beta^-\gamma$, $t_{½}$ 30.0 y) is being used to study cesium chemistry it is necessary to wait only 15–20 min after sampling until counting as the daughter ^{137m}Ba (IT, $t_{½}$ 2.55 min) reaches an equilibrium level within that time. Since the ratio of the ^{137m}Ba and the ^{137}Cs activity is the same in all samples at equilibrium, the total count rate before and after a chemical step is then a true measure of the behavior of cesium alone. If radioactive equilibrium is not re-established in a convenient time, it may be necessary to either discriminate against the activity not involved in the chemical system, to take into account its contributions to the net count rate, or to remove it immediately before counting.

It may be necessary or expedient to use a radioactive nuclide which can undergo significant decay during the chemical investigation. In these cases, in order to compare results at different points in the process, it is necessary to correct all counts to the same time (usually starting time of the experiment).

18.2. CHEMISTRY OF TRACE CONCENTRATIONS

Consider a sample containing a pure radionuclide with a disintegration rate of 10^7 dpm. For a $t_{½}$ of 1 h the number of atoms is (§5.11) 8.7×10^8; for a $t_{½}$ of 1 y it is 7.6×10^{12}. If such a sample is dissolved in one liter of solution, the respective concentrations would be 1.4×10^{-15} M and 1.3×10^{-11} M. At such concentrations the chemical behaviour may